Cloud Computing Notes

Sample Study Material for RAG Testing

Introduction to Cloud Computing

Cloud computing is the delivery of computing resources such as servers, storage, databases, networking, and software over the internet.

Benefits

Major benefits of cloud computing include scalability, flexibility, reduced upfront infrastructure costs, availability, and access to resources on demand.

Service Models

Infrastructure as a Service, or IaaS, provides virtualized computing resources. Platform as a Service, or PaaS, provides a platform for developing and deploying applications. Software as a Service, or SaaS, provides complete software applications over the internet.

Deployment Models

Public cloud resources are provided over the internet for multiple customers. Private cloud resources are dedicated to one organization. Hybrid cloud combines public and private cloud environments.

Virtualization

Virtualization allows multiple virtual machines to run on a single physical machine. It improves resource utilization and flexibility.

Scalability

Scalability is the ability of a system to handle increasing workloads by adding or adjusting resources.

Cloud Security

Cloud security includes protecting data, applications, identities, and infrastructure. Common practices include strong authentication, encryption, access control, and backups.

Example

An organization can deploy an application on cloud infrastructure and increase computing resources when the number of users grows.